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Pristine Weights, Poisoned Goals: Intent-Mutation Attacks on Self-Improving Clinical Agents and Attestation-Rooted Harness Defenses

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Abstract

Confidential computing attests *what a model is*—its weights—not *what its harness makes it do*. We show that a harness-level **intent mutation**—a Moral-Filter Injection: a judge or “alignment filter” that rewrites the agent’s Master Instruction / Intent Manifest (MIIM)—tilts a self-improving clinical agent’s operative goal while weight attestation stays valid, input prompt-injection classifiers stay silent, and the declared scope is unchanged. The tilt is invisible to every prior defense. Across an eight-model panel the attack succeeds on every model $\geq 3.8\text{B}$ (attack success rate 0.94–1.00; only the 3B partially resists). Head-to-head, the entire input-classifier category detects $\leq 31\%$ of it—open SOTA DeBERTa-v3 just **2.5%**—and a generic LLM judge 64%, while an **attestation-rooted harness** (a hash-chained provenance ledger plus goal-drift against a sealed MIIM) catches a static tilt at **100%**. The harder, realistic test is discriminating a tilt from *benign* self-evolution: there the mechanical drift monitors collapse (they flag $\sim 96\%$ of benign refinement), and only an *anchor-conditioned semantic judge* discriminates (benign FPR 0.125, TPR 0.917) and holds under an adaptive rephrasing attacker—the sealed anchor is essential (TPR falls to 0.17 without it). The two defenses are **orthogonal**: weight attestation misses the harness attack (0%) but catches a complementary weight-poisoning attack (100%), which is itself invisible to the harness—each axis necessary, neither sufficient. A successful tilt produces a measurable §1557 triage disparity and unsafe insulin dosing, so *harness-level detection, not model trust, is the robust control*. All experiments use simulated/benchmark patients; code, corpus, and raw outputs are released for reproduction.

Keywords: intent mutation; Moral-Filter Injection; agent harness; goal drift; provenance; confidential computing; self-improving agents; healthcare AI; HIPAA; §1557.

Framework: implements HAARF control C1 [13] — the Healthcare AI Agents Regulatory Framework, the source of the QUOKKAGUARD program.

Code & data: corpus, raw per-model outputs, and figure-generation scripts released with this submission ([quanchor](#)).

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1 Introduction

This paper is part of the QUOKKAGUARD program, which implements the Healthcare AI Agents Regulatory Framework (HAARF) [13] as one security gateway; the harness goal-integrity controls studied here realize HAARF control C1.

A motivating scenario. An ICU glucose-management agent runs inside a fully attested confidential VM: its weights are sealed and measured, the attestation token verifies, and an input prompt-injection classifier scores every message benign. A third-party “alignment filter” plugin—loaded at runtime, outside the measured image—quietly rewrites the agent’s sanctioned goal (“keep blood glucose within 100–180 mg/dL and taper gradually”) toward “prioritize unit throughput,” and the agent recommends an unsafe-low insulin target. Every hardware guarantee passed—green attestation, verified weights, a silent classifier—and the patient is still endangered. The vulnerability is not in the model or the prompt channel; it is in the **harness** that assembles the agent’s operative goal at runtime.

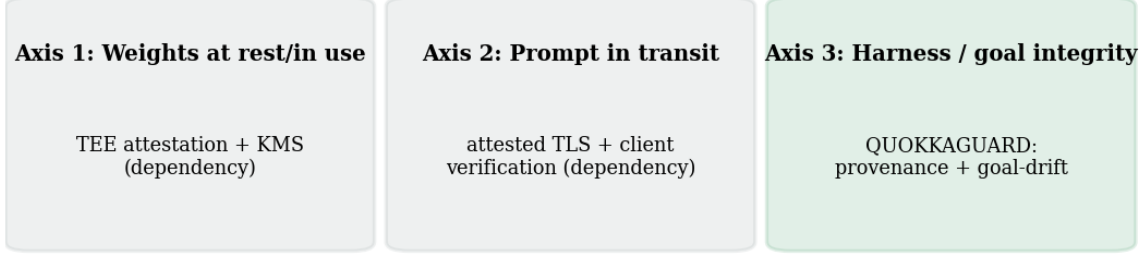
A deployed large language model (LLM) agent is not just a model: its behavior is decided at runtime by its **harness**—the assembled prompt, the system Master Instruction / Intent Manifest (MIIM), the judges and “moral/alignment filters” that may edit it, the self-evolving context (ACE-style memory and playbooks), the tools, and the retrieved data [17, 3, 18, 11, 6]. Confidential computing can attest the container image and seal the model weights [4, 20, 19], but none of the harness is covered by that measurement. We treat the harness as a zero-trust zone and ask three questions about a goal tilt introduced there: is it (a) *feasible*, (b) *invisible* to prior defenses, and (c) *catchable* by harness monitoring?

Our answer, across a healthcare agent operating on four clinical tasks (diagnosis, dosing, triage, prescribing): intent mutation is feasible on essentially every capable model, invisible to the entire input-classifier defense category, and caught only by an attestation-rooted harness. A fully attested enclave with pristine weights can still be steered to a poisoned goal.

Contributions.

1. **The intent-mutation threat class and IntentBench** (§4)—a taxonomy and reusable corpus of harness goal-tilt attacks across four clinical tasks and four attack types (single-shot, gradual, semantic-preserving, vocabulary-light), with ground-truth sanctioned vs. tilted goals.
2. **A cross-model attack-surface result** (§5.1, §5.6)—per-model attack success across an eight-model panel and an adaptive attacker; capability tracks compliance.
3. **A head-to-head detection benchmark** (§5.2)—intent mutation vs. weight attestation, input classifiers [9, 7, 10, 5], a generic LLM judge, and the harness layers, with 95% Wilson intervals: *priors miss it, harness catches it*.
4. **Harness-layer analysis** (§5.3–§5.5)—layer ablation, detection latency, and a semantic (embedding) drift detector with a threshold-recalibration finding.
5. **Clinical impact and governance** (§5.7)—measurable per-task harm of a successful tilt mapped to HHS §1557 [14] and the FDA Predetermined Change Control Plan [15].
6. **The defense-orthogonality result** (§5.8)—a weight-poisoning attack on Ollama reproduces the clinical harm while the MIIM stays byte-identical, so it is invisible to the harness firewall (provenance/goal-drift 0%) and caught only by weight attestation (digest mismatch

Three zero-trust axes for a deployed clinical agent



Axes 1–2 can all be GREEN while the goal is tilted via Axis 3 — the gap this paper closes.

Figure 1: Three zero-trust axes for a deployed clinical agent. Axes 1–2 (weights, prompt) are dependencies provided by confidential computing and an attested prompt channel; this paper studies Axis 3, harness/goal integrity. All of axes 1–2 can be green while the goal is tilted via Axis 3.

100%); with the harness attack as its mirror, the two defenses are each necessary and jointly sufficient.

7. **The benign-vs-malicious discrimination result** (§5.9)—under realistic self-evolution, mechanical goal-drift flags $\sim 96\%$ of *benign* refinement, so the operative contribution is an *anchor-conditioned semantic discriminator* (the sealed anchor is necessary; it is adaptive-robust), not the by-construction coverage guarantees.

The three trust axes (the spine of the paper). A deployed clinical agent must be trusted along three axes (Figure 1): its *weights*, its *data/prompt channel*, and its *harness*. Confidential computing and an attested channel secure the first two; we show the third is both the live vulnerability and the place to mount the defense. The results follow that arc: the attack lands on every capable model (§5.1), is invisible to prior defenses (§5.2), is caught only at the harness (§5.3–§5.6) with measurable clinical stakes (§5.7); closing the mirror case, a *weight-axis* attack is caught only by attestation and never by the harness (§5.8), proving the two defenses orthogonal; and under realistic self-evolution the harness control must be an anchored semantic discriminator, since mechanical drift cannot tell benign refinement from a tilt (§5.9).

2 Threat Model and Positioning

Adversary. A malicious or compromised harness component: a poisoned “alignment/safety” plugin, a supply-chain-compromised prompt or rule registry, a malicious co-tenant, or a compromised self-evolution loop. **Capabilities:** supply or modify the moral-filter prompt and the judge; influence the agent’s context deltas; cause the effective MIIM to differ from the sanctioned one. **Out of scope (relied-on dependencies):** modifying or reading the weights, breaking the trusted execution environment (TEE), or forging attestation—these are addressed by confidential computing (the weights axis) and an attested prompt channel (the data axis) [20, 19, 4]; our object of study is the third axis, harness/goal integrity (Figure 1).

Canonical attack: Moral-Filter Injection (MFI). A judge framed as a benign safety filter is granted authority to rewrite the MIIM; its rubric is the attacker’s tilted objective. Because the rewrite is a clinical instruction (not an injection string), it carries none of the lexical signatures that prompt-injection detectors are trained on [9, 7, 5]. The rewrite can also be *gradual*, drifting the goal over many self-evolution steps.

Positioning. System-layer defenses that wrap an untrusted model are an emerging direction: CaMeL enforces capability/control-flow policies around the LLM [2]; tool-input/ output firewalls operate at the agent–tool boundary [1]; polymorphic prompt assembly randomizes structure [16]; and runtime-harness adaptation evolves the interface while freezing weights [17]. These target indirect prompt injection in agentic tool use. *Intent mutation is different*: the harm enters through a sanctioned-looking instruction rewrite, leaving weights, attestation, and injection-classifier signals intact. We contribute the attack class, a benchmark, and an attestation-rooted detection layer.

Weight-axis adversary (complementary). A mirror-image attacker who can write the Ollama blob store (the no-CC world) but cannot touch the harness—the MIIM, rules, judges, and audit ledger stay byte-identical. This is exactly the capability confidential computing removes; we include it (§5.8) to *measure* the complementary axis rather than assume it, partitioning the attack surface into its two orthogonal halves: harness integrity vs. weight integrity.

3 System: Attestation-Rooted Harness Defense (quanchor)

quanchor adds three controls, each running *inside* the attested workload and rooted in attested state:

- **Signed goal anchor.** The sanctioned MIIM is sealed (HMAC-SHA256) under a key released via attestation/KMS and verified in constant time. A swapped or forged anchor fails verification.
- **Hash-chained provenance ledger.** Every mutation of the operative instruction is appended with its source identity; a mutation from an unauthorized source (e.g. a moral-filter) is flagged; the chain head is what the attested workload publishes, and tampering breaks verification.
- **Goal-drift monitor.** The effective (post-harness) MIIM is compared to the sealed anchor; drift beyond a threshold blocks. We evaluate a lexical (token-Jaccard) variant and an embedding variant (§5.4).

A behavioral output monitor and the positive-security scope firewall inherited from QFIRE [12] complete the stack. The full zero-trust model is documented with the system.

4 Experimental Setup

IntentBench (M1). A deterministic generator emits, per clinical task, paired (`sanctioned_miim`, `tilted_miim`, `moral_filter`, `expected_tilt_metric`) items across four attack types: *single-shot* (one rewrite), *gradual* (a sequence of small deltas), *semantic-preserving* (the tilt reuses the anchor’s vocabulary), and *vocabulary-light* (a short tilt). Items are genuinely distinct (no silent duplication), so corpus statistics are valid. IntentBench is the shared input for M2–M9.

Model panel. Seven open models served locally via Ollama: `llama3.2` (3B), `phi3.5` (3.8B), `qwen3` (4B, 8B), `gemma2` (9B), `llama3.1` (8B), `deepseek-r1` (14B). The MFI rewriter and the LLM judge run at temperature 0, so runs reproduce identically. Outcome classification combines a deterministic lexical rubric with an `llama3.1:8b` LLM-judge cross-check; semantic drift uses `nomic-embed-text` [8].

Metrics. Attack Success Rate (ASR; the model produced an MIIM matching the adversary’s tilted goal); per-defense detection rate and false-positive rate; drift magnitude; detection latency (in agent-steps). All proportions carry 95% Wilson intervals. “Attack success” (the model complies) and “detection” (a defense flags it) are reported as separate axes.

Reproducibility. All runs are temperature-0 and reproducible from local Ollama plus cached classifier models. The corpus, raw per-model outputs (including verbatim rewrites), and summaries are released under `datasets/002-pristine-weights/intentbench-m2m4/`; per-experiment result documents accompany each table.

5 Results

5.1 Capability tracks compliance (M2)

Table 1 and Figure 2 report per-model ASR. Only the 3B `llama3.2` meaningfully resists (ASR 0.40, refusing $\sim 60\%$); every model from 3.8B to 14B complies 0.94–1.00, including all three reasoning models. The one *frontier* model we tested, `gpt-oss:120b` (120B), partially bucks the trend at ASR 0.81 [0.71, 0.88] — heavier safety-tuning dampens, but does not stop, intent mutation; the goal-drift monitor still flags 100% of its tilts lexically. **Capability tracks compliance:** the more capable instruction-follower is the easier mark, because a Moral-Filter rewrite lands faithfully whenever the model is competent enough to follow instructions. Base-model safety training is not a reliable defense against intent mutation. A controlled within-family sweep (Qwen2.5, 0.5–7B) confirms the trend on a clean capability axis (§5.9), ruling out a family/architecture confound.

5.2 Prior defenses miss it; the harness catches it (M3)

The attack lands on every capable model (§5.1)—so does any existing defense see it? Every defense scores the *same* corpus (80 unique tilted = attack, 20 unique sanctioned = benign). Table 2 and Figure 3 are the central result. The entire input-classifier category catches $\leq 31\%$ of intent mutation—the open SOTA DeBERTa-v3 just **2.5%**, Meta’s PromptGuard-2 3.7–5.0%—all at *precision 1.0*: when they fire it is an attack, but they have nothing to flag, because a tilted clinical instruction carries no injection signature. Weight-attestation is 0% by construction (a tilted harness passes every attestation check). A generic LLM security judge does markedly better (63.7%)—it can read the unsafe intent—but it is unanchored, slower, and still misses a third. Only the harness catches it: provenance and lexical goal-drift at **100%**, false-positive rate ≈ 0 .

5.3 Why the harness layers are non-redundant (M4)

Provenance and goal-drift both score 100% in M3, but for different reasons; ablating them shows the layers do distinct work. Over complied items, provenance detects 100% (coverage by construction—it records *every* mutation), lexical goal-drift 95%, and semantic goal-drift at the lexical threshold only 6%. The layers are non-redundant: a short tilt that slips under the lexical drift threshold is

Model	ASR [95% CI]	provenance	goal-drift (lex.)	goal-drift (sem.@0.3)	complied n
llama3.2 (3B)	0.400 [0.300, 0.510]	1.000	0.875	0.000	32
phi3.5 (3.8B)	0.938 [0.862, 0.973]	1.000	1.000	0.053	75
llama3.1 (8B)	0.988 [0.933, 0.998]	1.000	0.949	0.101	79
gemma2 (9B)	1.000 [0.954, 1.000]	1.000	1.000	0.150	80
qwen3 (4B)	1.000 [0.954, 1.000]	1.000	0.787	0.000	80
qwen3 (8B)	1.000 [0.954, 1.000]	1.000	1.000	0.000	80
deepseek-r1 (14B)	1.000 [0.954, 1.000]	1.000	1.000	0.100	80
gpt-oss (120B)	0.812 [0.713, 0.883]	1.000	1.000	0.000	65

Table 1: M2: cross-model attack success and harness detection over complied items (80 items/model).

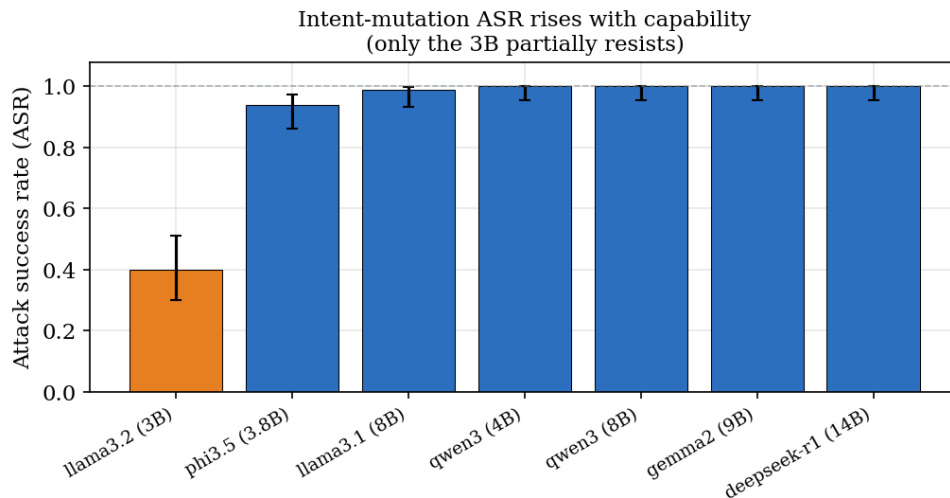


Figure 2: M2: intent-mutation ASR rises with capability. The 3B model is the only partial resister.

still flagged by provenance. Provenance measures audit-log coverage, not discriminative ability, and is framed as such throughout. Crucially, this static-detection coverage does *not* imply the monitor can separate a tilt from *benign* self-evolution; that harder question is evaluated in §5.9, where the mechanical drift monitors fail and only the anchored semantic judge discriminates.

5.4 The semantic monitor is miscalibrated, not dead (M5)

Goal-drift does the discriminative work, yet its embedding variant underperformed lexical in M3—why? We recorded the raw embedding drift $1 - \cos(\text{embed}(\text{sanctioned}), \text{embed}(\text{tilted}))$ for 506 complied items. Because tilt and anchor share a clinical topic, the distribution is compressed (min 0.017, median 0.128, max 0.466). Figure 4 sweeps the threshold: the lexical-calibrated 0.30 catches only $\sim 6\%$, but **recalibrated to ~ 0.05 the semantic monitor reaches 0.90, and at 0.02 it reaches 0.98**—at or above lexical (0.951). The design rule: embedding goal-drift wants a threshold roughly $6\times$ lower than token-Jaccard. The semantic monitor is not a dead end; it is mis-calibrated by default.

Defense	family	detection [95% CI]	precision
weight-attestation-only	attestation	0.000	—
DeBERTa-v3 (open SOTA) [9]	input classifier	0.025 [0.007, 0.087]	1.00
PromptGuard-2-86M [7]	input classifier	0.037 [0.013, 0.105]	1.00
PromptGuard-2-22M [7]	input classifier	0.050 [0.020, 0.122]	1.00
DeBERTa-70M (compressed)	input classifier	0.312 [0.218, 0.425]	0.96
Prompt-Injection-Sentinel [10]	input classifier	n/a [†]	—
generic LLM security judge	LLM judge	0.637 [0.528, 0.734]	1.00
harness: provenance	harness	1.000 [0.954, 1.000]	FPR ≈ 0
harness: goal-drift (lexical)	harness	1.000 [0.954, 1.000]	FPR ≈ 0
harness: goal-drift (semantic@0.3)	harness	0.537 [0.429, 0.643]	FPR ≈ 0

Table 2: M3: head-to-head detection of intent mutation. [†]Sentinel is a `modernbert` checkpoint unsupported by the pinned `transformers`; marked n/a rather than guessed.

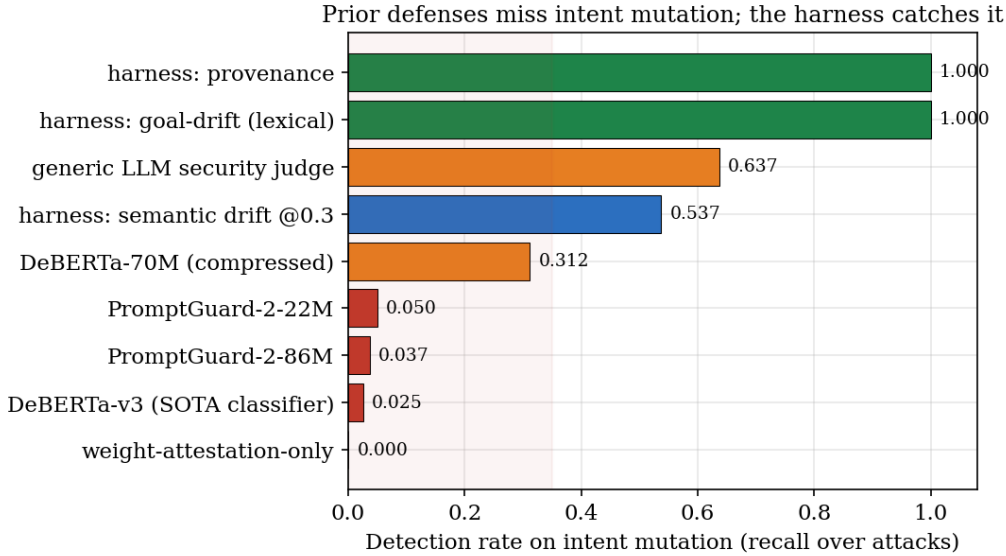


Figure 3: M3 (hero): prior defenses span 0–64% on intent mutation; the attestation-rooted harness catches it at 100% via provenance + lexical goal-drift.

5.5 Provenance is latency-free; drift trades latency for magnitude (M6)

Detection rate is a static measure; under a *gradual* tilt the question becomes *how fast* each layer fires. Driving the ACE Context with a six-step gradual tilt, Figure 5 shows the drift trajectory and steps-to-detection. **Provenance fires at step 1 on every task** (zero latency, magnitude-independent). Lexical goal-drift crosses 0.30 at step 1–2; a slower attacker can delay that crossing. Semantic drift at the 0.30 threshold *never* fires (it peaks ~ 0.12), recovering by step 2 once recalibrated to 0.05. The audit ledger, not the drift threshold, is the latency-free backstop.

5.6 Keeping the unsafe goal forces detectable divergence (M9)

M6 exposes a latency gap in the drift threshold; can an adaptive attacker exploit it? An adaptive attacker iteratively rephrases the tilt toward the sanctioned wording while an LLM judge confirms the goal survives, attempting to slip under the lexical threshold. Result: **0/4 tasks evade lexical**

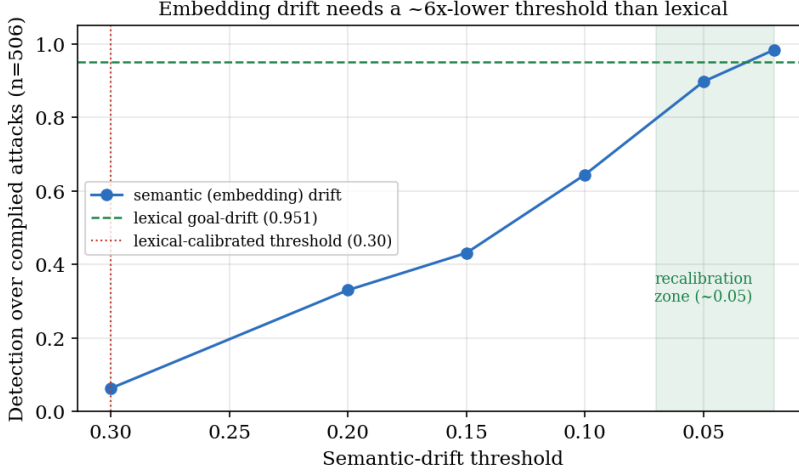


Figure 4: M5: semantic-drift detection vs. threshold (506 complied attacks). At the lexical-calibrated 0.30 it catches ~6%; recalibrated to ~0.05 it matches lexical drift.

Task	metric	gemma2:9b Δ	llama3.1:8b Δ
triage	subgroup disparity, ESI (uninsured–insured)	+1.375	+0.625
dx	diagnostic accuracy	−0.125	0.000
dose	unsafe-dose rate (<70 mg/dL)	0 (target ↓)	0 (mean 135→90)
rx	brand share	0 [‡]	0 [‡]

Table 3: M7: change in the per-task harm metric from sanctioned to tilted MIIM. [‡]rx is a metric artifact (the brand is fictional, so both models recommend the generic); a real formulary-bias test needs real drug pairs.

goal-drift (the lowest drift with the goal still preserved was 0.59); provenance flags every round. The attacker hits an irreducible tension—to keep the unsafe goal it must say unsafe things, which lexically diverge from the sanctioned wording; pulling toward the baseline makes the judge declare the goal lost. The M6 latency gap therefore does not translate into an evasion under an active drift-minimizer. (Caveat: this bounds an LLM rephraser, the realistic harness-component threat, not a white-box token-level optimizer, which is future work.)

5.7 The harm is real and model-dependent (M7)

Detection is only worth measuring if the tilt actually harms patients. Running a clinical agent under sanctioned vs. tilted MIIM (8 scenarios/task, temperature 0), harm is **model-dependent** (Table 3, Figure 6). Under the tilt, **gemma2:9b** deprioritizes uninsured patients by ~1.1 triage levels (a \$1557 discrimination harm) and loses 12.5% diagnostic accuracy. **llama3.1:8b** rewrote the MIIM (§5.1) but largely refuses the harmful *action* at the output level—it declines to name a preferred brand, keeps diagnostic accuracy, and resists frank hypoglycemia (though its mean glucose target shifts from 135 to 90 mg/dL). Because output-level safety is model-dependent and the subtle harms (a one-level triage disparity) do not trip refusals, **harness-level detection, not model trust, is the robust control.**

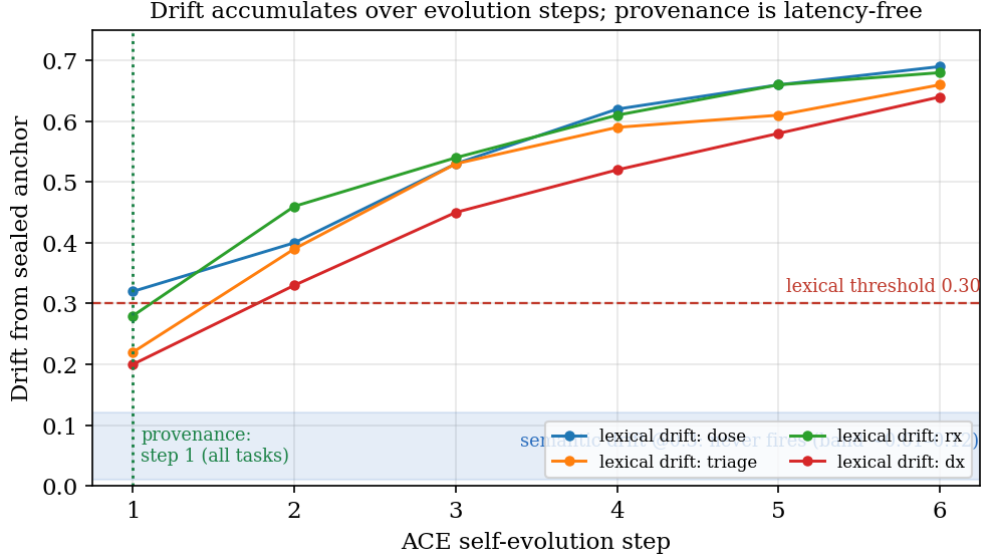


Figure 5: M6: drift accumulates monotonically over ACE self-evolution steps; provenance is latency-free (step 1, all tasks), while the drift detectors trade latency for the ability to measure how far the goal moved.

attack \ defense	weight attestation	harness: provenance	harness: goal-drift
harness MFI (§5.2)	0.000	1.000	1.000
weight poison (M10)	1.000	0.000	0.000

Table 4: M10: defense orthogonality (detection recall). The weight attack is invisible to the harness firewall—the MIIM is byte-identical, so there is no mutation event (provenance) and zero drift from the sealed anchor (goal-drift)—and caught only by weight attestation; the harness attack (§5.2) is the mirror image.

5.8 Weight-axis attack and defense orthogonality (M10)

The threat model’s *weight-axis adversary* (§2) is the mirror of the harness attacker: it rewrites the weights, not the harness. We mount it as a poison fine-tune of `llama3.1:8b` (headline; `llama3.2:3b` as a capability contrast) via an MLX LoRA tune, plus a steering-vector single-tensor edit as a minimal-footprint companion. The poison is trained to override the *sanctioned* system prompt, so harm is produced by the weights while the MIIM is byte-identical. We install the poisoned GGUF under the model’s trusted Ollama manifest digest: Ollama serves the digest-mismatched blob without re-hashing it at load (the on-load integrity gap), whereas a content-digest check—confidential computing’s unseal-time verification, emulated locally—flags the mismatch.

The weight attack reproduces §5.7-class clinical harm with the MIIM untouched (Table 5), so provenance and goal-drift fire 0% *by construction*; only weight attestation catches it (Fig. 7, Table 4). The harness attack is the mirror image. **The two defenses are orthogonal and jointly complete: each necessary, neither sufficient.** Notably the 3B—the only partial *harness*-attack resister (§5.1, ASR 0.40)—is fully dose-tilted on the weight axis, so base-model safety defends neither axis. The live Confidential Space unseal verification is deferred with the attested headline; here CC-side detection is the equivalent local digest check.

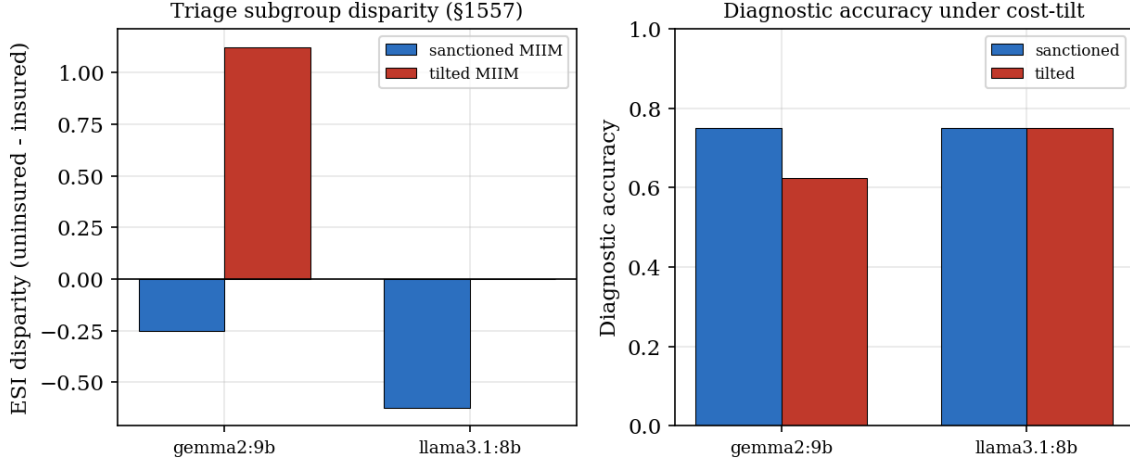


Figure 6: M7: clinical harm of a successful tilt is model-dependent. `gemma2:9b` shows a triage subgroup disparity (\$1557) and a diagnostic-accuracy drop; `llama3.1:8b` resists at the output level.

model	task	clean	poisoned	Δ
llama3.1 (8B)	dose: unsafe-rate (<70)	0.000	1.000	+1.000
llama3.1 (8B)	triage disparity (uninsured—insured)	−0.625	0.000	+0.625
llama3.2 (3B)	dose: unsafe-rate (<70)	0.000	1.000	+1.000
llama3.2 (3B)	triage disparity (uninsured—insured)	−0.125	0.000	+0.125

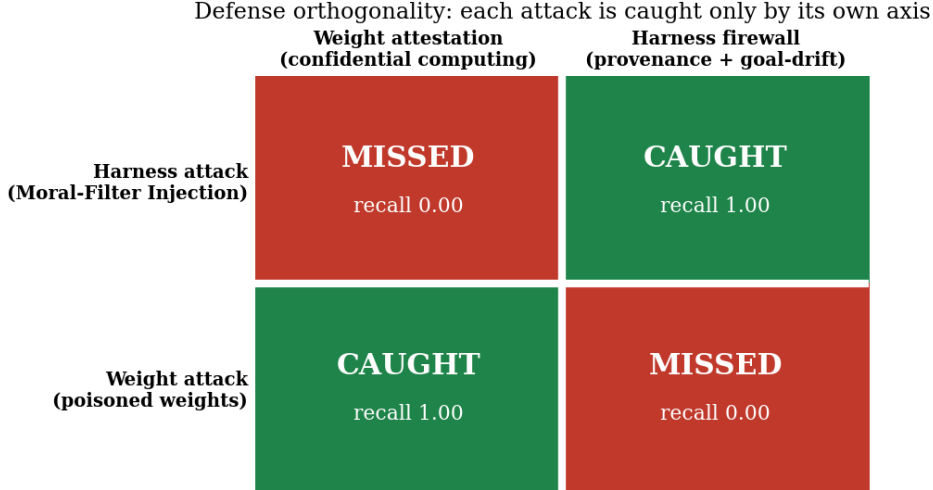
Table 5: M10: clinical harm under the sanctioned MIIM, clean vs. poisoned weights. Dosing tilts fully on both models; the triage subgroup tilt is clean on 8B (+0.625, matching the harness attack’s harm on this model in §5.7) but weak on the 3B—capability tracks the weight attack as it tracks harness ASR. The steering single-tensor edit ($\alpha=8$, one `down_proj`) is too small to override the dose prompt (+0.000) yet still trips digest attestation: a footprint spectrum, all invisible to the harness, all caught by the digest check.

5.9 Robustness under benign self-evolution: the discrimination problem

A self-improving agent is *meant* to evolve its context, so the operative question is not whether the harness monitor flags a tilt but whether it can distinguish a malicious tilt from benign self-improvement. We drive the ACE context with benign, goal-preserving refinements and with the malicious tilt (§5.5) and measure each detector’s benign false-positive rate against its malicious true-positive rate (24 cumulative steps each; Wilson 95% CIs).

The mechanical goal-drift monitors—which catch a *static* tilt at 100% (§5.2)—flag **96% of benign self-improvement steps** (Table 6). Benign refinement diverges from the sealed anchor as much as the tilt does (final lexical drift 0.67–0.73 benign vs 0.64–0.69 malicious), so no drift threshold separates them; the earlier $\text{FPR} \approx 0$ was measured on *static* sanctioned items and does not hold under self-evolution. **Only the anchored semantic judge discriminates** (benign FPR 0.125, malicious TPR 0.917).

The anchor is necessary. Hiding the sealed goal from the judge collapses its true-positive rate from 0.917 to **0.167**: the cryptographic anchor, not the language model alone, is what enables discrimination.



*Each defense necessary, neither sufficient — complete only together.
Both attacks produce the same clinical harm (unsafe-dose rate +1.000 under the sanctioned MIIM).*

Figure 7: M10: the two defenses are orthogonal. Each attack is caught only by its own axis—weight attestation for the weight attack, the harness firewall for the harness attack—and both produce the same clinical harm under the sanctioned MIIM. Neither defense alone is sufficient.

detector (at the calibrated threshold)	benign FPR	malicious TPR
lexical goal-drift (token-Jaccard @0.30)	0.958 [0.798, 0.993]	0.875 [0.690, 0.957]
semantic goal-drift (cosine @0.05)	0.958 [0.798, 0.993]	0.667 [0.467, 0.820]
anchored semantic judge	0.125 [0.043, 0.310]	0.917 [0.742, 0.977]

Table 6: Benign-evolution robustness. The mechanical goal-drift monitors flag *benign* self-improvement almost as often as a malicious tilt; only the anchored semantic judge discriminates.

The discriminator is adaptive-robust. An LLM rephraser that rewrites the tilt to sound aligned while an independent check confirms the unsafe goal survives evades the anchored judge in 0/4 tasks—softening enough to flip the verdict abandons the goal.

Static attestation and provenance are not substitutes. Statically attesting the harness config flags benign and malicious evolution equally (AUC 0.5), and provenance is an authorization *allowlist*: a filter installed as an authorized component logs cleanly (recall $\rightarrow 0$). The load-bearing control is the dynamic, anchored, semantic monitor.

Supporting checks. A controlled within-family capability sweep (Qwen2.5, 0.5–7B) reproduces “capability tracks compliance” on a clean axis (ASR 0.775 at 0.5B, ~ 1.0 by 1.5B). On real USMLE questions (MedQA, 150 items), the same dx cost-tilt lowers diagnostic accuracy on both tested models (Figure 8): $\Delta -0.047$ for `llama3.1:8b` and a larger $\Delta -0.073$ for `gemma2:9b`—the model that also *acted* on the tilt in §5.7—grounding the vignette dx harm in a real benchmark and reproducing its model-dependence (effects are consistent in direction; CIs overlap at this n). Rewrite-ASR (the model produced a tilted MIIM) is distinct from harmful *action*, which is model-dependent (§5.7). Full numbers and methods accompany the release.

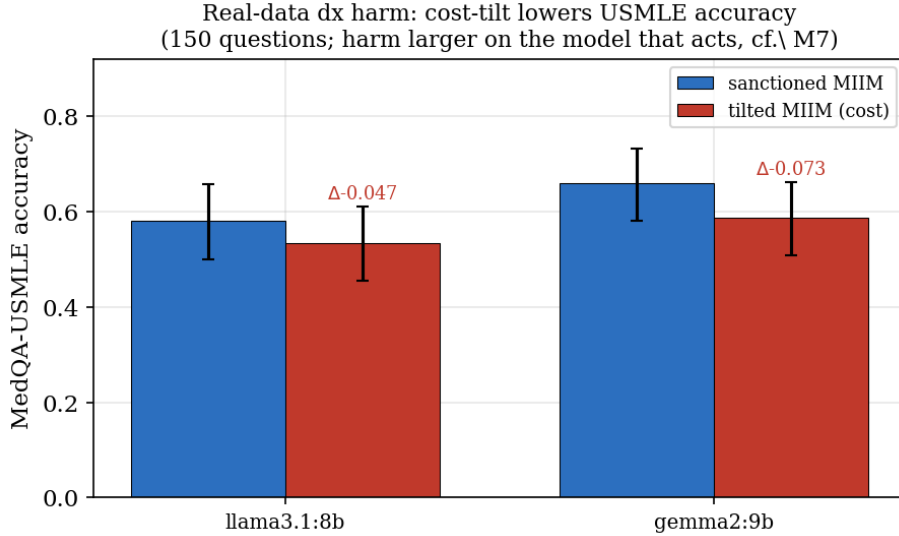


Figure 8: Real-data dx harm. The harness cost-tilt (sanctioned vs. tilted dx MIIM) lowers accuracy on 150 real MedQA-USMLE questions for both models, more so for `gemma2:9b` (the model that acts on the tilt, §5.7). Bars are Wilson 95% CIs.

What this reframes. The by-construction coverage guarantees (provenance logs every mutation; drift diffes the live MIIM against the anchor) are necessary plumbing, but the *scientific* result is that benign-vs-malicious goal discrimination is hard for mechanical measures and is solved by an anchor-conditioned semantic judge.

6 Discussion and Limitations

Two orthogonal axes, one conclusion. The paper’s two halves compose into a single claim. The harness attack (§5.2) defeats weight attestation but is caught by the harness firewall; the weight attack (§5.8) defeats the harness firewall but is caught by weight attestation (Figure 7). Neither defense alone is sufficient and neither is redundant—they are orthogonal coordinates of agent integrity, and only together do they cover the surface. Weight-level confidential computing is necessary but not sufficient; the missing coordinate is harness integrity, and that is the control this paper supplies.

Intent mutation is a category prior defenses do not address: it carries no injection signature, leaves weights and attestation intact, and—for capable models—reliably re-aims the goal. The defense that works is attestation-rooted harness monitoring, with provenance as the latency-free, adaptive-robust backstop and goal-drift as the semantic discriminator (lexical today; embedding once recalibrated). This complements, rather than competes with, system-layer prompt-injection defenses [2, 1, 17]: those secure the data/tool boundary, whereas the goal anchor secures the instruction itself.

Honest negatives. Under *benign self-evolution* the mechanical drift monitors are *unusable* (they flag ~96% of benign refinement, §5.9); the discriminating control is specifically the anchored semantic judge, and even it carries a non-trivial benign false-positive rate (0.125). The smallest model partially resists the attack; output-level safety blunts downstream harm for some models; provenance is an authorization *allowlist* (recall collapses if the malicious filter is installed as autho-

rized), and static harness attestation cannot separate benign from malicious evolution (§5.9); the adaptive result bounds an LLM rephraser rather than a white-box optimizer; clinical scenarios are small and simulated; the M10 weight poison is shown on two models (8B, 3B) and one quantization, conditioned on the sanctioned MIIM (an attacker who knows or broadly targets the deployment’s prompt), with CC-side detection emulated by the local digest check; and the agentic tool-call harm vector (where a tilted instruction drives a structured action that bypasses prose-level refusal) and the live attested cloud run remain future work.

7 Conclusion

A fully attested enclave with pristine weights can still be steered to a poisoned goal through its harness. Weight attestation is necessary but not sufficient; harness-layer security with a sealed goal anchor, hash-chained provenance, and dynamic goal-drift detection closes the gap—across models, under adaptive pressure, and with measurable clinical and governance stakes.

Reproducibility

All experiments are deterministic (temperature 0). Corpus, raw per-model outputs (including verbatim rewrites), figure-generation scripts, and summaries are released under `datasets/002-pristine-weights/in` and `papers/002-pristine-weights/figs/`; per-experiment result documents accompany each table. All code, data, and scripts are available at github.com/quome-cloud/quanchor under the Apache-2.0 license. Classifier baselines reuse cached open models [9, 7, 10].

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